Stochastic Simulation (MIE1613H) - Homework 2

Due: March 3, 2026

- Submit your homework on Quercus as a single PDF file (with complete answers and explanations) together with the Python source codes for each question requiring coding (.py files) or a single Jupyter Notebook (.ipynb file).
- At the top of your homework include your **name**, **student number**, **department**, and **program**.
- Late submissions will be penalized 10% each day the homework is late.
- You may discuss the assignment with other students in general terms, but each student must solve the problems, write the code and the solutions individually.
- Fix the random seed in all your simulation code using `np.random.seed()` or `SimRNG.InitializeRNSeed()` in PythonSim at the beginning of your code. The reported results in your submission must match with the output of your code on our end.
- Refer to the course syllabus for guidelines on permissible use of Generative AI. Note that all AI-generated content must be properly cited, and you will not receive credit for code / content you cannot fully explain.
- **It is very important to explain your answers (including the logic of your simulation code and estimation approach) clearly and not just provide the code/results. Full marks will only be given to correct solutions that are fully and clearly explained.**

Problem 1. **(15 Pts.)** Let $X \sim \text{Poisson}(\lambda)$ with pmf

$$p_k = P(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots$$

Suppose we wish to generate X using acceptance–rejection. As a proposal distribution, consider a geometric random variable Y on $\{0, 1, 2, \ldots\}$ with pmf

$$q_k = (1 - \rho)\rho^k, \qquad 0 < \rho < 1,$$

which is easy to generate samples from using the inversion method. Consider the following algorithm:

1. Generate Y with $P(Y = k) = q_k$.
2. Generate $U \sim \text{Unif}(0, 1)$ independently.
3. If,
$$U \leq \frac{p_Y}{c\, q_Y},$$
accept and return $X = Y$; otherwise reject and repeat.

(a) Find a finite constant c such that
$$p_k \leq c\, q_k \qquad \text{for all } k \geq 0.$$

(b) Show that with such a constant c, the algorithm returns a random variable having the Poisson(λ) distribution. **Hint**: You need to show that $P(Y = k | U \leq p_Y/cq_Y) = P(X = k)$ using the same approach we used for continuous random variables in the lecture.

(c) Show that the probability of accepting a proposal in any trial is $1/c$, and hence the expected number of trials required to generate one observation is c.

Problem 2. **(25 Pts.)** Consider a single bike share station over the course of a day. The station has capacity C docks and initially contains $B(0)$ bikes ($0 \leq B(0) \leq C$). Customers arrive to *rent* bikes and bikes arrive to be *returned* throughout the day.

The attached datasets (`BikeShare_Trips_Start.csv` and `BikeShare_Trips_End.csv`) provide the time-stamps of all trips in December 2016 that respectively started from and ended in station Broadway & W 60 St.

Consider the following stochastic model of the station. Rental requests follow a nonstationary Poisson process (NHPP) with time-varying rate $\lambda_d(t)$, and returns follow an independent NHPP with time-varying rate $\lambda_r(t)$, for $0 \leq t \leq 24$.

When a rental request occurs at time t:

- If $B(t) \geq 1$, the customer rents a bike immediately and $B(t^+) = B(t) - 1$.
- If $B(t) = 0$, the customer leaves immediately (an *unhappy* customer) and $B(t^+) = 0$.

When a return occurs at time t:

- If $B(t) \leq C - 1$, the bike is returned and $B(t^+) = B(t) + 1$.
- If $B(t) = C$, the station is full and the return is rejected (an *unhappy* customer) and $B(t^+) = C$.

(a) Partition the day into K equal-length time bins (e.g., $K = 24$ one-hour bins or $K = 48$ half-hour bins). Using the n days of data, estimate $\lambda_d(t)$ and $\lambda_r(t)$ as *piecewise-constant* rates:
$$\hat{\lambda}_d(t) = \hat{\lambda}_{d,k} \quad \text{and} \quad \hat{\lambda}_r(t) = \hat{\lambda}_{r,k} \qquad \text{for } t \text{ in bin } k.$$

(b) Using the method discussed in lecture, examine whether a Poisson process model is appropriate for: (i) rental requests and (ii) returns.

(c) Explain how to simulate rental-request events and return events over $[0, 24]$ hours using the thinning algorithm with the piecewise-constant rate estimates from part (a).

(d) Develop a discrete-event simulation model in `PythonSim` to simulate sample paths of $B(t)$ (number of cars in the station) over one day. **Note**: To receive full mark, you need to provide an explanation of the elements of your discrete-event simulation, i.e., events, how they impact the state, and what future events would need to be scheduled at each event similar to the examples from the lecture and textbook.

(e) Let U be the total number of unhappy customers over a day (lost rentals + rejected returns). Using your simulation model estimate $E[U]$ and construct a 95% confidence interval for $E[U]$. Assume $B(0) = 25$ and $C = 40$.

Problem 3. **(25 Pts.)** A company would like to decide how many items of a certain product it should keep in inventory for each of the next T months (T is a fixed input parameter). The times between demands are IID exponential random variables with a mean of 0.1 month. The sizes of the demands, D, are IID random variables (independent of when the demands occur), where

$$D = \begin{cases} 1 & \text{w.p. } 1/6 \\ 2 & \text{w.p. } 1/3 \\ 3 & \text{w.p. } 1/3 \\ 4 & \text{w.p. } 1/6 \end{cases}$$

where w.p. is short for "with probability."

At the beginning of each month, the company reviews the inventory level and decides how many items to order from its supplier. If the company orders Z items, it incurs a cost of $K + iZ$, where $K = \$32$ is the setup cost and $i = \$3$ is the incremental cost per item ordered. (If $Z = 0$, no cost is incurred.) When an order is placed, the time required for it to arrive (or **lead time**) is a random variable that is exponentially distributed with mean 0.75 month. Different orders have independent lead times, and multiple orders may be outstanding at the same time.

The company uses an (s, S) policy to decide how many to order, i.e.,

$$Z = \begin{cases} S - IP, & \text{if } IP < s, \\ 0 & \text{if } IP \geq s, \end{cases}$$

where IP is the **inventory position**, defined as the current inventory on hand plus the total number of items already on order but not yet received.

When a demand occurs, it is satisfied immediately if the inventory level is at least as large as the demand. If the demand exceeds the inventory level, the excess demand is lost. In this case, only the available inventory is sold and the inventory level becomes zero.

Let $I(t)$ be the inventory level (on-hand inventory) at time t (note that $I(t) \geq 0$ for all t). A sample path of $I(t)$ is a piecewise-constant function that decreases at demand epochs and increases when orders arrive.

Assume that the company incurs a holding cost of $h = \$1$ per item per month held in inventory. Since $I(t)$ is the number of items held at time t, the time-average (per month) inventory level over the T-month period is

$$\bar{I} = \frac{\int_0^T I(t)\,dt}{T},$$

so the average holding cost per month is $h\bar{I}$.

If demand is lost, the company incurs a penalty of $\ell = \$12$ per unit of unsatisfied demand, representing lost profit and goodwill. Let $L(t)$ denote the number of lost demand up to time t, then average lost demand per month is,

$$\bar{L} = \frac{L(T)}{T},$$

so the average lost sales cost is $\ell\bar{L}$.

Assume that the initial inventory level is $I(0) = 50$ and that no order is outstanding. We are interested in comparing the following six inventory policies with respect to their performance over $T = 120$ months:

s	20	20	20	30	30	30
S	50	60	70	60	80	100

(a) For each of the following outputs, determine whether they are a discrete-time output or a continuous-time output of the simulation: inventory level, lost demand, and ordering cost.

(b) Develop a discrete-event simulation model in `PythonSim` to estimate the performance of a given (s, S) policy. **Note**: To receive full mark, you need to provide an explanation of the elements of your discrete-event simulation, i.e., events, how they impact the state, and what future events would need to be scheduled at each event similar to the examples from the lecture and textbook.

(c) For each policy, estimate (with a 95% confidence interval) and report the **expected average total cost per month**, defined as the sum of average ordering cost, average holding cost, and average lost sales cost.

Note: As we will discuss later in the course you should use Common Random Numbers (CRN) when comparing alternative scenarios. That is, use the same random numbers for different scenarios. This can be done easily by re-calling the method `SimRNG.ZRNG = SimRNG.InitializeRNSeed()` in `PythonSim` before evaluating the next scenario.

Problem 4. (15 Pts.) The attached files (`APPL.csv` and `SQ.csv`) provides daily stock prices for Apple and Square in 2023.

(a) Fit two independent Geometric Brownian Motions (GBM) to the **Adj Close** price for the two stocks by estimating the drift μ and volatility terms σ for each stock. Recall that for GBM the log-returns are independent Normal random variables, i.e.,

$$\log\left(\frac{S(t_i)}{S(t_{i-1})}\right) \sim N(\mu(t_i - t_{i-1}), \sigma(t_i - t_{i-1})),$$

for all $t_i, i = 1, \ldots, k$.

(b) Are the log-returns well-described using the fitted independent Normal random variables in each case? Use your estimates from part (a) together with both graphical methods and statistical goodness-of-fit tests discussed in lecture to justify your answer.

(c) Extend the model to a bi-variate GBM (with correlated prices) by estimating the covariance matrix of the log-returns for the two stocks.

Problem 5. (20 Pts.) A Laplace (double-exponential) distribution with location μ and scale $b > 0$ has CDF

$$F(x) = \begin{cases} \frac{1}{2}\exp\left(\frac{x-\mu}{b}\right), & x < \mu, \\ 1 - \frac{1}{2}\exp\left(-\frac{x-\mu}{b}\right), & x \geq \mu. \end{cases}$$

(a) Propose an inversion algorithm to generate i.i.d. samples from Laplace(μ, b) using $U \sim$ Unif$(0, 1)$.

(b) Derive the theoretical quartiles, i.e., the 0.25th quantile $Q(0.25)$ and the 0.75th quantile $Q(0.75)$ as functions of (μ, b), and show that the interquartile range satisfies

$$\text{IQR} = Q(0.75) - Q(0.25) = 2b \ln 2.$$

Hint: The inverse CDF function that you obtained in part (a) should be useful for finding the theoretical quantiles.

(c) A matching estimator fits parameters by matching the sample median and the sample interquartile range:

$$\hat{\mu} = \text{median}(x_1, \ldots, x_n), \qquad \hat{b} = \frac{\widehat{\text{IQR}}}{2 \ln 2}.$$

Generate $n = 1000$ i.i.d. samples from Laplace(μ, b) with $\mu = 0$ and $b = 2$ using part (a). Compute $(\hat{\mu}, \hat{b})$ and compare them to the true parameters.

(d) Construct a 95% bootstrap confidence interval for μ and b based on the $n = 1000$ sample from part (c) using $B = 1000$ bootstrap resamples.

1. For $r = 1, \ldots, B$, draw a bootstrap sample $(x_1^{*(r)}, \ldots, x_n^{*(r)})$ by sampling with replacement from the original data.
2. Compute bootstrap estimates $(\hat{\mu}^{*(r)}, \hat{b}^{*(r)})$ using the same matching formulas as in part (c).
3. Let $\hat{\mu}^*_{(0.025)}$ and $\hat{\mu}^*_{(0.975)}$ be the 0.25th and 0.975th quantiles of $\{\hat{\mu}^{*(r)}\}_{r=1}^B$. Report the CI $[\hat{\mu}^*_{(0.025)}, \hat{\mu}^*_{(0.975)}]$. Do the same for b.

State whether the true values $\mu = 0$ and $b = 2$ fall inside your intervals.